

EXPERIMENT – 2 (GO FURTHER – 1)

PROJECT NAME

BODMAS Method Based Calculator

OBJECTIVE

To design a calculator using Arduino UNO, OLED display, and keypad that performs arithmetic calculations according to the BODMAS rule.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
SSD1306 OLED Display	1
5×4 Matrix Keypad	1
Breadboard	1
Jumper Wires	Required

PIN CONNECTIONS

OLED

OLED Pin	Arduino Pin
SDA	A4
SCL	A5

Keypad

Keypad Pin	Arduino Pin
R1–R5	D2–D6
C1–C4	D7–D10

KEYPAD FUNCTIONS

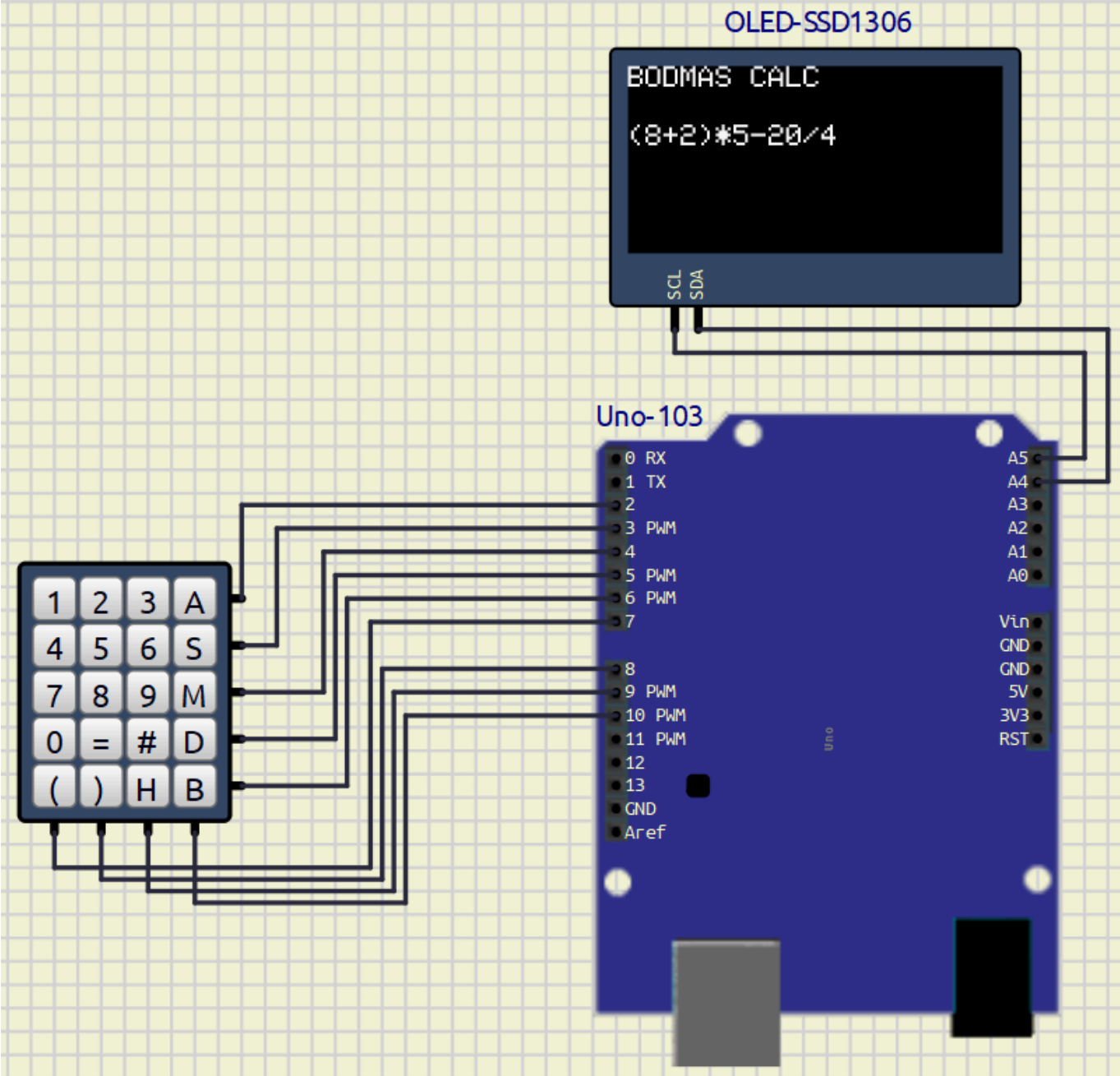
Key	Function
A	+

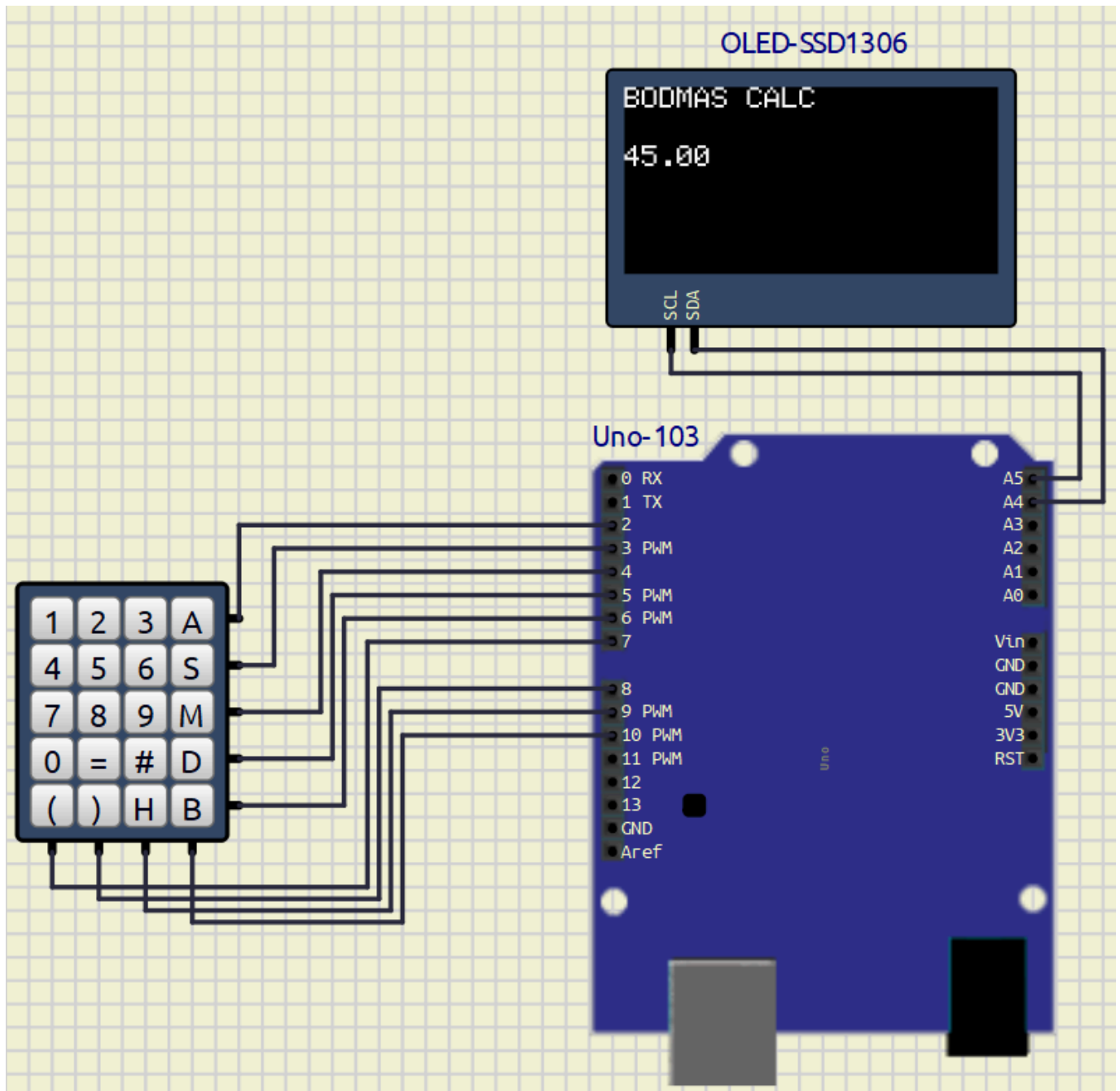
Key	Function
S	-
M	*
D	/
=	Calculate
#	Clear
B	Backspace
H	History

SOFTWARE USED

- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM





MAIN ARDUINO CODE

```
//=====
// BODMAS Calculator using Arduino UNO
// OLED Display + 5x4 Keypad
//
// Features:
// - Addition
// - Subtraction
// - Multiplication
// - Division
// - Brackets
// - History
// - Backspace
// - BODMAS Rule
```

```
//=====

//===== LIBRARIES =====

// I2C Communication Library
#include <Wire.h>

// OLED Graphics Library
#include <Adafruit_GFX.h>

// OLED SSD1306 Library
#include <Adafruit_SSD1306.h>

// Matrix Keypad Library
#include <Keypad.h>

//===== OLED SETTINGS =====

// OLED width
#define SCREEN_WIDTH 128

// OLED height
#define SCREEN_HEIGHT 64

// Create OLED display object
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

//===== KEYPAD SETTINGS =====

// Number of rows in keypad
const byte ROWS = 5;

// Number of columns in keypad
const byte COLS = 4;

// Keypad button layout
char keys[ROWS][COLS] = {

  {'1','2','3','A'},
  {'4','5','6','S'},
  {'7','8','9','M'},
  {'0','=',' #','D'},
  {'(',' )','H','B'}
};

// Arduino pins connected to keypad rows
byte rowPins[ROWS] = {2,3,4,5,6};

// Arduino pins connected to keypad columns
```

```
byte colPins[COLS] = {7,8,9,10};

// Create keypad object
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

//===== VARIABLES =====

// Stores current mathematical expression
String expression = "";

// Pointer used by parser
const char* expr;

// Stores last 5 calculations
String history[5];

// Current history index
int historyIndex = 0;

//===== OLED DISPLAY FUNCTION =====

// Function to refresh OLED screen
void refreshDisplay() {

    // Clear OLED screen
    display.clearDisplay();

    // Set text size
    display.setTextSize(1);

    // Set text color
    display.setTextColor(SSD1306_WHITE);

    // Title
    display.setCursor(0,0);
    display.print("BODMAS CALC");

    // Expression display position
    display.setCursor(0,20);

    // Display current expression
    display.print(expression);

    // Small delay for stable Proteus simulation
    delay(50);

    // Show everything on OLED
    display.display();
}
```

```
//===== PARSER FUNCTIONS =====

// Skip blank spaces
void skipSpaces() {

    while(*expr == ' ')
        expr++;
}

// Forward declaration
double parseExpression();

//===== NUMBER PARSER =====

// Reads numbers from expression
double parseNumber() {

    double number = 0;

    // Continue while digit found
    while(isdigit(*expr)) {

        // Build number digit by digit
        number = number * 10 + (*expr - '0');

        expr++;
    }

    return number;
}

//===== FACTOR PARSER =====

// Handles brackets
double parseFactor() {

    // Ignore spaces
    skipSpaces();

    // Check for opening bracket
    if(*expr == '(') {

        // Move to next character
        expr++;

        // Solve expression inside bracket
        double result = parseExpression();

        // Skip closing bracket
        if(*expr == ')')
```

```
        expr++;

        return result;
    }

    // Otherwise parse normal number
    return parseNumber();
}

//===== TERM PARSER =====

// Handles multiplication and division
double parseTerm() {

    // Read first factor
    double result = parseFactor();

    while(true) {

        skipSpaces();

        // Multiplication
        if(*expr == '*') {

            expr++;

            result *= parseFactor();
        }

        // Division
        else if(*expr == '/') {

            expr++;

            result /= parseFactor();
        }

        else {
            break;
        }
    }

    return result;
}

//===== EXPRESSION PARSER =====

// Handles addition and subtraction
double parseExpression() {

    // Read first term
    double result = parseTerm();
```



```
while(true) {

    skipSpaces();

    // Addition
    if(*expr == '+') {

        expr++;

        result += parseTerm();
    }

    // Subtraction
    else if(*expr == '-') {

        expr++;

        result -= parseTerm();
    }

    else {
        break;
    }
}

return result;
}

//===================================================== EVALUATE FUNCTION =====

// Evaluates full mathematical expression
double evaluate(String s) {

    // Convert String to character pointer
    expr = s.c_str();

    // Start parsing expression
    return parseExpression();
}

//===================================================== SETUP =====

void setup() {

    // Start serial monitor
    Serial.begin(9600);

    // Initialize OLED
    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

    // Clear OLED
```

```
display.clearDisplay();

// Text size
display.setTextSize(1);

// Text color
display.setTextColor(SSD1306_WHITE);

// Cursor position
display.setCursor(0,20);

// Startup message
display.print("READY");

// Show message
display.display();

// Wait 2 seconds
delay(2000);

// Show empty calculator screen
refreshDisplay();
}

//===== MAIN LOOP =====

void loop() {

    // Read keypad button
    char key = keypad.getKey();

    // If any key pressed
    if(key) {

        // Print pressed key on serial monitor
        Serial.print("Key: ");
        Serial.println(key);

        //===== CALCULATE =====

        // If "=" pressed
        if(key == '=') {

            // Evaluate expression
            double result = evaluate(expression);

            // Save calculation in history
            history[historyIndex] =
            expression + "=" + String(result);

            // Move to next history slot
            historyIndex++;
        }
    }
}
```

```
// Reset history index
if(historyIndex >= 5) {

    historyIndex = 0;
}

// Replace expression with result
expression = String(result);

// Update OLED
refreshDisplay();
}

//===== CLEAR SCREEN =====

// If "#" pressed
else if(key == '#') {

    // Clear expression
    expression = "";

    // Refresh OLED
    refreshDisplay();
}

//===== BACKSPACE =====

// If "B" pressed
else if(key == 'B') {

    // Remove last character
    if(expression.length() > 0) {

        expression.remove(expression.length()-1);
    }

    // Refresh OLED
    refreshDisplay();
}

//===== HISTORY =====

// If "H" pressed
else if(key == 'H') {

    // Clear OLED
    display.clearDisplay();

    // OLED settings
    display.setTextSize(1);
```

```
display.setTextColor(SSD1306_WHITE);

// Title
display.setCursor(0,0);
display.print("HISTORY");

int y = 12;

// Show saved history
for(int i=0; i<5; i++) {

    if(history[i] != "") {

        display.setCursor(0,y);

        display.print(history[i]);

        y += 10;
    }
}

// Show on OLED
display.display();

// Wait 4 seconds
delay(4000);

// Return to calculator screen
refreshDisplay();
}

//===== NORMAL INPUT =====

else {

    // Convert keypad symbols

    if(key == 'A')
        expression += '+';

    else if(key == 'S')
        expression += '-';

    else if(key == 'M')
        expression += '*';

    else if(key == 'D')
        expression += '/';

    // Add normal characters
    else
        expression += key;
```

```
    // Refresh OLED
    refreshDisplay();
  }
}
```

WORKING PRINCIPLE

1. User enters expression using keypad.
2. Arduino receives keypad input.
3. OLED displays the expression.
4. On pressing =, Arduino evaluates the expression using BODMAS rule.
5. Result is displayed on OLED.
6. Previous calculations are stored in history.

OUTPUT / RESULT

Example

Input:

$(8+2)*5-20/4$

Output:

45

The calculator successfully performed calculations according to the BODMAS rule.

ADVANTAGES

- Supports BODMAS calculations
- Compact and portable
- Real-time OLED display
- History and backspace support

APPLICATIONS

- Educational projects
- Scientific calculator systems
- Arduino learning

CONCLUSION

The BODMAS calculator using Arduino UNO was successfully implemented. The calculator correctly evaluated mathematical expressions and displayed the results on the OLED display.

REFERENCES

1. <https://www.arduino.cc/>
2. <https://docs.arduino.cc/hardware/uno-rev3/>
3. https://github.com/adafruit/Adafruit_SSD1306
4. <https://playground.arduino.cc/Code/Keypad/>
5. <https://www.circuitbasics.com/how-to-set-up-a-keypad-on-an-arduino/>